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Computational Evaluation of Mel4 and Lact
Interactions with Adenovirus and Norovi
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INTRODUCTION
Human adenovirus and norovirus are environmentally stable, non-enveloped viruses that remain major causes of infections worldwide.
Effective antiviral therapeutics remain limited.
We have developed a novel antimicrobial peptide, Mel4 1
Mel4 and lactoferricin (LFc) possess antiviral activity against adenovirus type 5 (HAdV-5) and murine norovirus type 1 (MNV-1) 2
At active concentrations they are not toxic to the host mammalian cells. 2
AIM
This study aims to determine the mechanism of action of Mel4 and LFc using electron microscopy and in silico approaches.
MATERIALS and METHODS
Human Adenovirus type 5 (HAdV-5; ATCC-VR-5) and murine norovirus type 1 (MNV-1; ATCC-VR-1937)
Mel4 (KNKRKRRRRRRGGRRRR) and LFc (RRWQWRMKKLE)
Viruses exposed to the peptide/mimics for 2 h at 37°C. Stained with 2% phosphotungstic acid. Examined using transmission electron microscopy.
For HAdV-5, the hexon, penton, and fiber proteins were obtained, while for MNV-1, the capsid protein VP1 was included.
Structural models were predicted using AlphaFold3
Structures were visualized and examined using PyMOL (Schrodinger; https://www.pymol.org) and Discovery Studio.
Docking used HDOCK (http://hdock.phys.hust.edu.cn/)
GROMACS software package was used to perform molecular dynamics simulations on a katana GPU-enabled platform.
The particle mesh Ewald (PME) approach was used to determine long-range electrostatic interactions, and a cutoff distance of 1.2 nm was used for van der Waals interactions.
HAdV-5
MNV-1
HAdV-5 + Mel4
MNV-1 + Mel4
HAdV-5, Mel4 had IC50 of 47.4 µM but LFc had no activity. Mel4 targeted the capsid, releasing electron dense material
MNV-1, Mel4 had IC50 of 8.6 µM, LFc had IC50 of 23.18 µM. Mel4 also resulted in release of electron dense material
Figure 1: Comparative docking score
capsid proteins.
Blue = Mel4
Red = LFc
Docking Score
-280
-270
-260
-250
-240
-230
-220
-210
HAdV-5 - Hexon
HAdV-5 - Penton
HAdV-5 - Fiber
MNV-1 – VP-1
Figure 2: Residue level interaction of Mel4 with
Table 1:
Peptide
Mel4
Lactoferricin
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